

Examiner reports

Q1.

Parts (a) and (b), about the assumptions, were generally answered well. The most common incorrect answer was to say that the peg was fixed in part (a).

There were many good solutions to part (c). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach “lost” a negative sign in their working and ended up with a positive acceleration.

In part (d) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an acceleration of 9.8 instead of 0.98.

Q2.

There were many correct force diagrams on the scripts, but some candidates lost marks due to missing arrows or poor labelling. The printed answer in part (b) seemed to help the candidates as there were many correct answers. It was clear that in some cases candidates had needed to change their approach to obtain the correct value. There were also many good answers to parts (c) and (d), but some candidates included extra forces which meant that their friction value was wrong. A few candidates seemed to confuse the coefficient of friction and the friction force.

Q3.

Part (a) of this question was done very well, but part (b) caused difficulties for some candidates. The most common errors in part (b) were to include the weight of the car and trailer or to produce equations in which the wrong combinations of forces were used.

Q4.

There were lots of good responses to this question, although part (e) was sometimes incorrect.

Part (a) was found to be very straightforward and the majority of the candidates gained the marks, even if they did not go on to do other parts of the question.

Part (b) was also done well, with the vast majority of candidates stating clearly the two equations of motion that they went on to solve. A few did use a method with only one equation, but fortunately this method was not common. Also, a few used the weight of the block instead of the friction force.

Part (c) was also done well, provided the candidates had an appropriate equation to use from part (b).

It was interesting that some candidates who had found parts (a) to (c) difficult were able to use the acceleration given and produce correct answers to parts (d) and (e). There were many good solutions to part (d), but in part (e) candidates often continued to use the

acceleration as 1.225 m s^{-2} or took the initial velocity of the particle to be zero. Some candidates made both of these errors. The success rate for part (e) was lower than that for part (d).

Q5.

The candidates found part (a) of this question relatively straightforward, but part (b) was found to be very demanding, and very few correct solutions were seen. Candidates lost marks in part (a) mainly due to errors in the resolving: for example using $\sin 30^\circ$ instead of $\cos 30^\circ$ in part (a)(ii). A few candidates lost one mark because they gave their final answer for the coefficient of friction as 0.31 rather than 0.313.

In part (b), very few candidates realised that they needed to express the normal reaction in terms of the tension. This error resulted in solutions that were flawed. Most of the candidates who did express the normal reaction in terms of the tension went on to gain full marks for this part of the question, although a few did make arithmetic or algebraic errors.

Q6.

Most candidates did well on this question, having few difficulties finding the magnitude of the friction force. In part (b), the great majority of candidates took a correct approach to finding the acceleration. There were a few candidates who did not consider the block and the particle as two separate bodies. Part (c) was done well with candidates experiencing very few difficulties.

Q7.

This question caused more difficulties. There were a large number of poor force diagrams. The most common errors were to include a reaction force and to show the two tensions as equal, by using the same letter to indicate the magnitude of each tension.

In part (b), there were candidates who did not show enough working to obtain full marks. Reasons for this included not showing a trigonometric term such as $\sin 30^\circ$ or $\cos 60^\circ$ and not showing how the value of 78.4 was obtained from the candidate's initial equation. A few candidates tried to resolve the weight instead of the tension.

The candidates did find part (c) easier, with some candidates only obtaining marks on this part of the question, using the printed result in part (b).

Q8.

Generally part (a) was done well, whilst part (b) caused more difficulties. The fact that the acceleration was given did seem to help many candidates, but some did not justify the minus sign. Applying the constant acceleration equation to find the distance was also done well, although a few candidates took $u = 0$ and $v = 4$. There were a number of candidates who used two constant acceleration equations, finding the time taken to stop as an intermediate step. A few of these candidates did not go on to find the distance.

In part (b), many of the candidates were able to find the magnitude of the friction force. In a few cases the candidates did not show how the reaction force was obtained. Full marks were not awarded if the candidates simply stated $R = 1.84$ without any justification. When calculating the acceleration there were many sign errors in candidates' equations and some candidates also included a " 4 " which appeared as if it was a force. While many candidates realised that the puck would remain at rest, only a relatively small number of

them were able to give a convincing explanation. Many of the explanations contained vague statements and lacked precision and clarity.

Q9.

Part (a) was done very well and some candidates had clearly been helped by the presence of the printed answer. There were several examples of candidates who obtained the printed answer in part (b) showing enough supporting working. A value of 4000 was often introduced without any working to show that it had been obtained from an application of Newton's Second Law.

Several candidates found the tension in the rope first and then found the value of P . Some candidates were confused about when to include the tension in the rope. The responses to part (c) were variable, but those who had taken a systematic approach to part (b) usually did well. The answers for part (d) were very varied, with some very good explanations and other responses which lacked coherence.

Q10.

The candidates made good attempts at this question. Some found it difficult to get going in part (a). Most of the candidates who had difficulties had tried to consider particle A instead of particle B . Generally the candidates made good attempts at parts (b) and (c). In part (d), the main problems were for candidates who found forming an equation of motion difficult. Often, these candidates had also found part (a) difficult. Most candidates were able to make an attempt at part (e) using their answers to parts (c) and (d).

Q11.

This question was also done well by many candidates. In part (a), there were some confused responses and occasional problems with minus signs. Part (b) was also generally done well. In part (c), a few candidates had problems. The most common of these was to omit an arrow from the diagram to show the force; the other was to use the

lengths incorrectly when calculating the angle, for example $\tan^{-1}\left(\frac{8}{2}\right)$ and $\sin^{-1}\left(\frac{8}{8.25}\right)$.

Q12.

This question caused the candidates more difficulty. A relatively small number were able to produce completely correct solutions. The majority found part (a) difficult to deal with, but were often able to pick up marks with correct solutions for the more familiar part (b). Many of the candidates did not make coherent attempts at part (a) and produced confused working.

Q13.

The force diagram in part (a) was often drawn well, but the usual errors of extra forces and missing labels or arrows were seen. Part (b) was done very well, with the printed answer helping almost all of the candidates to gain full marks. The responses to part (c) were very varied, with some solutions excellent and some very poor. One fairly common error was to form an equation with a sign error, like $F - 5g \sin 40^\circ = 5 \times 0.8$. Other more serious errors included using the normal reaction force instead of the weight component parallel to the slope, and simply assuming that the friction was equal to 4. Part (d) of the question was found to be very difficult, with very few candidates giving acceptable

answers. The majority of candidates stated that the coefficient of friction would increase.

Q14.

This question was also answered well by many candidates. A few lost a mark in part (a) by not showing that the two tensions were different, and a number of candidates used incorrect moments in part (b), but spuriously obtained the given answer by approximating

$$\frac{40g}{1.9} \text{ to } 21g.$$

Q15.

The candidates found this question demanding, especially part (a). The few candidates who considered the forces acting on the caravan had few difficulties and often found the correct force. Many candidates tried to consider the two bodies as a single particle and made little progress. Some candidates were able to do part (b), even if they had made little or no progress with part (a). As well as considering the forces on the wrong body, some candidates introduced a variety of sign errors into their equations. The real problem for the candidates in answering this question was that they did not recognise when to consider the two bodies separately and when to consider them as a single body.

Q16.

Candidates seemed to either do very well or very badly on this question, with many producing completely correct solutions. Some candidates did part (a) by finding the angle α first and then using trigonometry to find F , rather than finding F first as suggested in the question. There were some candidates who were able to form the two equations, $F \cos \alpha = 6$ and $F \sin \alpha = 5$ but who then made no progress in solving these to find either F or α . A small number of candidates did not know how to start this question and produced no working at all.

Q17.

Almost all candidates gained some marks on this question, but only a fairly small proportion gained full marks. In part (a), there were many good answers, but a number of candidates stated assumptions that were not about the string. Instead they produced comments about the block or the peg. Part (b) was generally done quite well, but there were some candidates who did not gain credit because they did not show enough working. For example, simply stating $6/4 = 1.5$ was not sufficient. There were, in contrast, some very good solutions that fully justified the final answer.

In part (c), the candidates who considered the forces on the 7 kg sphere did quite well. The main error that these candidates made was to use $T - 7g = 7a$, rather than $7g - T = 7a$ and obtained a tension of 79.1 N. Other candidates produced equations that involved the 13 kg block but could make little satisfactory progress because they did not know the friction force.

Part (d) was the most demanding part of the question. It required the candidates to find the friction force and then to use $F = \mu R$. Some candidates worked through these stages, but there were relatively few complete solutions. The most common errors were to assume that the friction was equal to the tension and simply apply $F = \mu R$, and to try to work with a combination of both bodies instead of just the block.

Q18.

In part (a) there were many good diagrams, but also a number that included extra forces, especially parallel to the slope. On some diagrams the arrows or labels were missing. Part (b) was done well and the vast majority of candidates did show enough working to gain all of the marks. Some candidates clearly used the given answer to correct their working. Part (c) was more demanding. When finding the acceleration, the main errors were not dividing the resultant force by the mass; making a sign error to obtain $a = -0.171$; not considering both forces, to produce equations like $100a = 68.4$; and not resolving the weight correctly.

In the last part, there were some good explanations, but there were also a lot that were not acceptable. Some candidates wrote about the resistance being related to the acceleration rather than the speed or velocity. A few also thought that the resistance would depend on the slope.

Q19.

Diagrams in part (a) were mostly good, with occasional errors in positioning the weight. Part (b) was mostly done correctly but there was some inaccuracy in manipulating simple equations to find the tensions. Most gave appropriate responses in part (c).

Q20.

This question was generally done very well, particularly parts (a) and (b). In part (c), some candidates simply assumed that the tension would be equal to the weight. Some of these candidates gave the same answer for part (d) and gained the mark there. A few candidates performed the calculation $4410 - 45$ and obtained an incorrect value.

Q21.

Parts (a) and (b) of this question were done well by many candidates, but part (c) was more demanding. In part (a) most candidates formed two equations as requested, but a few only worked with one equation for the whole system. Those who had done part (a) correctly had few problems with part (b), while those with incorrect equations in part (a) obtained incorrect answers as a result. The main problem in part (c) was that many candidates assumed that the particles had moved 1 m each, instead of the 0.5 m that was needed for them to be 1 m apart. A few candidates took the acceleration to be 9.8 m s^{-2} in this part of the question.

Q22.

This question, particularly part (a), proved to be more demanding and was not answered well by the weaker candidates. In part (a), some candidates ignored the instruction to resolve horizontally. The main issue was that the vast majority of candidates assumed that the tensions were equal and did not assign variables such as T_1 and T_2 to the tensions. In parts (b) and (c), some candidates did not take account of the fact that there were two tensions to take into account. In addition, some candidates made resolving errors, using $\sin 30^\circ$ instead of $\cos 30^\circ$. In part (c), some candidates felt that there was no need to resolve the forces, although they had done so in part (b). Also a small number of candidates found the weight rather than the mass.

Q23.

This question proved to be quite demanding for many candidates. In part (a), the main issue was that many candidates did not realise that they should consider the forces acting on the car. There were some who found two tensions, one based on the car and the other based on the truck. These solutions were often presented in such a way that they had to be treated as two attempts.

Many candidates were able to obtain the value of 600 in part (b), but some of the solutions did not contain any evidence of mechanics principles. If candidates were awarded full marks, they had to have demonstrated the consideration of the forces and an application of Newton's Second Law. Simply stating " $3000 - (800 + 1600) = 600$ " or similar was not considered sufficient.

Interestingly, there were many good answers to part (c), in some cases from candidates who scored no marks on (a) or (b). Some candidates were able to say that the tension would increase, but struggled to justify this.

Q24.

Many candidates were handicapped by the lack of a clear labelled diagram and so their moment equations were often incorrect. A few candidates forgot to include the g terms.

Q25.

Most candidates were successful in this question. The usual error occurred when candidates took moments about one end, normally the end on the land. They did not consider that the reaction vertically upwards on the rod was $65g$ N, acting at the point where the rod was potentially pivoting.

Q26.

Part (a) was often answered well but candidates appeared to find the requests complex and supplemented their answers with several lines of unnecessary working. Part (b)(i) proved popular and successful, but part (b)(ii) was very discriminating.

Q27.

A surprising number of diagrams in part (a) failed to include all the forces. Part (b) was very successful, showing a good understanding of the friction law. Part (c) was done well by many, but others couldn't appreciate the significance of the constant speed, the necessary starting point of the explanation. Many were successful in part (d), although some confused the normal reaction with the vertical component of the tension. Part (e) was generally done well with candidates gaining credit for correct calculations using their components of the tension force.

Q28.

Parts (a)(i), (ii), (v) and (b) proved routinely successful for well prepared and accurate candidates. Parts (a)(iii) and (iv) tested the understanding of the forces in the static situations and proved highly discriminating.

Q29.

Most candidates did very well on this question, although there were a few who did not

seem to be able to attempt this type of problem.

Q30.

Diagrams in part (a) were clear and mostly correct, with occasional marks lost due to missing arrows or an additional or incorrect force; the reaction force being vertical and the weight being given as 'g' were the most frequent errors. Part (b) was mostly done correctly, but part (c) was less successful, with extra forces sometimes present in equilibrium equations. The friction law was usually applied in part (d) but marks were lost through missing or incorrect inequality signs and rounding of answers to 2 significant figures.

Q31.

It was pleasing that most candidates formed, as requested in part (a), equations of motion for the two particles. Pleasingly very few used the 'whole approach instead of forming equations of motion for the two particles as requested. Most then proceeded to find the tension in the string. In part (b), rather than answer the question set, with a very small minority realising the necessity to consider the stone or the can in isolation.

Q32.

Again, this was answered very well, often scoring full marks.

Q33.

Diagrams in part (a) were clear and mostly correct, with occasional marks lost due to missing arrows or a missing or incorrect force, with the weight being given as 'g' as the most frequent. Parts (b) and parts (c) were also good, but few scored well in part (d), with one or even two forces missing from the equation of motion, the weight component being the most frequent omission.

Q34.

Again diagrams were mostly good, and parts (a)(ii) and (c) usually gained full marks. Part (b) was successfully completed by most candidates with the main error being the inclusion of the weight of P instead of the frictional force in the equation of motion of P. Pleasingly very few candidates used a 'whole string' approach instead of forming equations of motion for the two particles as requested. In part (d) the intended response focussed on the remaining force acting on each particle and the subsequent change in speed. Many candidates were successful here but others focussed on forces no longer present.

Q35.

There were some very good answers to this question, but a number of areas of difficulty were evident on some scripts. In part (a) a common error was to show the tension acting in an upward direction. Another error that appeared was to omit the normal reaction force. The vast majority of the candidates were able to produce the correct answer in part (b). Some candidates not find the tension directly, but found the reaction force first and then used this to find the tension. The fact that the tension was given helped some candidates, but there were a few solutions where not enough working was shown.

In part (c), one of the most common errors was to omit g from the moment equation, so that one of the terms was the product of a distance and a mass, rather than a distance

and a force. This often had an implication for the next stage when the candidates were finding the reaction force. Interestingly, some candidates took moments about the end of the rod, so that having an incorrect mass did not have an impact on their answer. Part (d) was generally done well.



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.